

Roll No.

TCS-527

B. TECH. (CS) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
APPLIED MACHINE LEARNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) For the given information, mean = 50, median = 55, standard deviation = 12, determine whether the coefficient of skewness is either positive, negative, or symmetric. (CO1)

OR

- (b) Explain the difference between supervised and unsupervised machine learning, with suitable examples. (CO1)
2. (a) Number of days during which nine members of staff did not use their cars : (CO2)

2 6 2 4 1 4 3 1 1

- (i) Determine Mean, Median and Mode for above.
(ii) Determine the interquartile range.

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(2)

OR

- (b) In a supermarket if the mean of expenditure of a shopper is 50 Euros, standard deviation is 15 and $n = 500$. Find the value below which 70% of shoppers are expected to spend. ($Q = 0.2000 \Rightarrow z = 0.524$). (CO2)
3. (a) Find the population variance and standard deviation of the age of children in a family of five children aged 16, 11, 9, 8, and 1. Also explain the condition where we apply central limit theorem. (CO1)

OR

- (b) If the petrol prices in UK is normally distributed with a mean of 1.30 Pounds and standard deviation is 0.08, determine the range where approximately 95% and 99% prices will fall. (CO1)
4. (a) Why Accuracy is not a complete performance measurement ? How you define kappa statistics. Explain with a scenario. (CO2)

OR

- (b) Explain the difference between : (CO2)
- (i) Precision and Recall
 - (ii) Sensitivity and Specificity
5. (a) In a Standard Normal Probability Distribution, how you calculate the Z-value. Also define the Z-score with a suitable example. (CO3)

OR

- (b) What is normal distribution ? Also explain the different shapes of probability distribution in the context of skewness. (CO3)